High-Dimensional Data Analysis Project

Group 3H

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1 Introduction (120 words)

The analysis of stock returns is a key task in finance, aimed at understanding the relationship between risk and return. This report analyses the monthly returns of 100 New York Stock Exchange listed companies from 2005 to 2019. Using high-dimensional data techniques, the study separates stock returns into systematic risk driven by overall market movements and idiosyncratic risk specific to individual firms or industries.

The objective is to examine whether latent factors align with observable market benchmarks such as the S&P 500. In particular, the analysis tests whether the first principal component represents the market, whether the Fama-French three-factor model explains common variation, which industries move with these factors, which industries carry higher systematic risk, and which industries may provide hedging benefits.

2 Data and Preparation (150 words)

Data preparation is the foundation of high-dimensional analysis. We utilize `SampleH.csv` (stock returns) and `MarketFactors.csv` (benchmarks).

Table 1: Sample of Dataset

year	date	stock_id	industry_letter	industry_name	return	market_return	smb	hml
2005	Y2005M1	F85464	F	Wholesale Trade	-0.028006	-2.76	-1.66	2.06
2005	Y2005M1	H89060	H	Finance, Insurance and Real Estate	-0.069402	-2.76	-1.66	2.06
2005	Y2005M1	H87476	H	Finance, Insurance and Real Estate	-0.040173	-2.76	-1.66	2.06
2005	Y2005M1	E79879	E	Transportation and Public Utilities	-0.321767	-2.76	-1.66	2.06
2005	Y2005M1	H10777	H	Finance, Insurance and Real Estate	-0.042159	-2.76	-1.66	2.06
2005	Y2005M1	B89901	B	Mining	0.154380	-2.76	-1.66	2.06

```
The dataset 'data_complete' contains:
- 14940 observations (rows)
- 9 variables (columns)
```

```
Total amoun of NA entries: 0
```

The final dataset contains 14,940 observations across 9 variables. This reflects the longitudinal nature of the data, covering multiple stocks over 180 months (January 2005 - December 2019). The data likely contained 83 stocks (14,940 / 180 months = 83).

The industry classification maps stocks to sectors such as Mining (B), Construction (C), Manufacturing (D), Transportation and Public Utilities (E), Wholesale Trade (F), Retail Trade (G), Finance, Insurance and Real Estate (H) and Services (I).

Variables include stock returns and three factors: Market Return, SMB (Small Minus Big, the size effect factor, capturing compensation for investing in small capitalization companies relative to large capitalization companies), and HML (High Minus Low, the value factor, capturing the compensation for value as measured by the book to market ratio).

3 Preliminary Analysis (EDA) (180 words)

Table 2: Group statistics by the 'industries'

industry_name	avg_return_monthly	volatility	num_stocks
Manufacturing	1.43%	12.87%	9
Wholesale Trade	1.23%	11.36%	6
Services	1.17%	16.48%	9
Mining	1.14%	15.95%	4
Transportation and Public Utilities	1.03%	12.25%	12
Retail Trade	0.96%	13.57%	10
Construction	0.80%	12.58%	1
Finance, Insurance and Real Estate	0.77%	7.29%	32

The table above summarises the performance and risk of eight industry sectors based on their monthly returns and volatility:

- **Top Performers**: Manufacturing leads with the highest average monthly return at 1.43%, followed closely by Wholesale Trade at 1.23%.
- **Risk and Volatility**: Services is the most volatile sector (16.48%), while Finance, Insurance and Real Estate is the least volatile (7.29%).
- **Sector Representation**: The Finance sector has the largest number of stocks in this dataset (32), whereas Construction is represented by only a single stock.

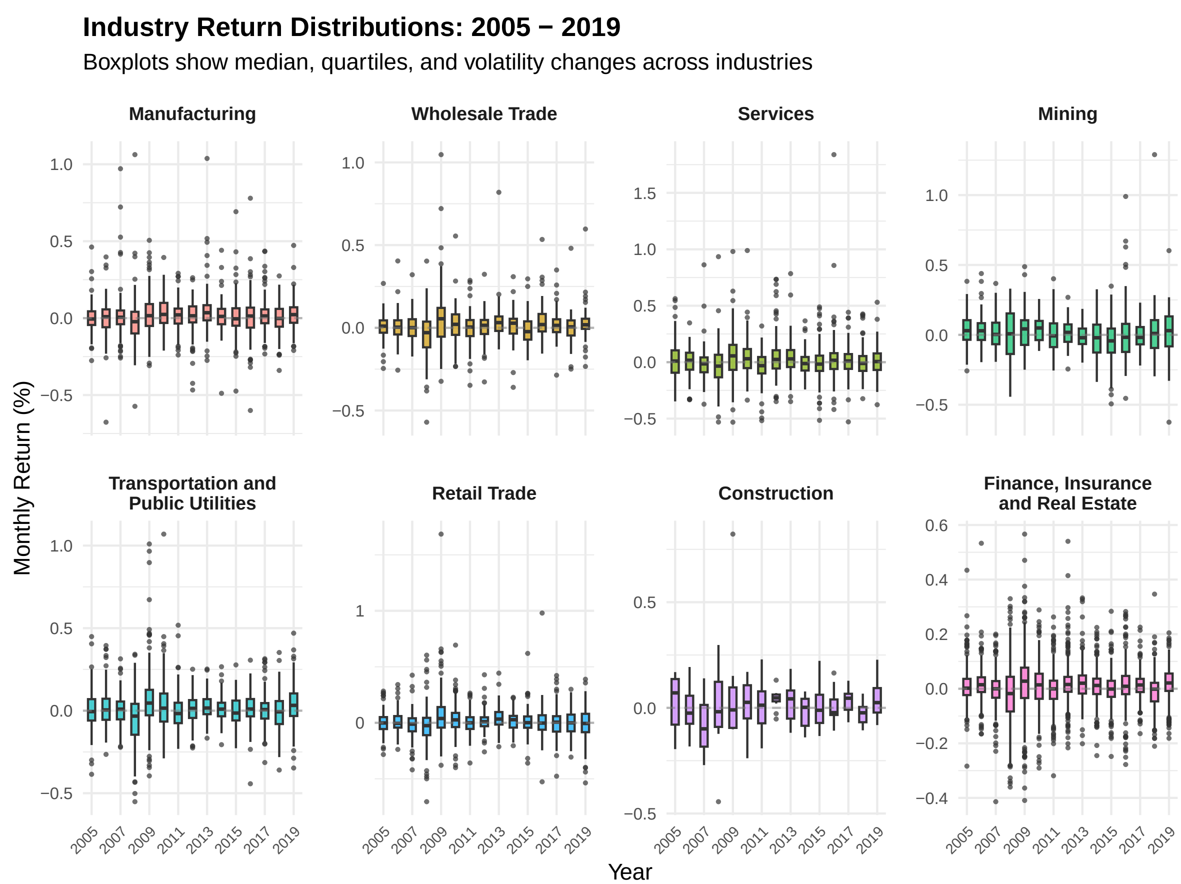

Figure 1: Distribution of monthly returns across industries from 2005 to 2019

As shown from the boxplots, we can see the industry return distributions from 2005 to 2019 reveal several key trends in market stability and sector performance:

- **Volatility Cluster from 2008 to 2009**: During this time, all industries saw a big rise in volatility. We can see this in the taller boxes (wider interquartile ranges) and the extreme outliers, especially in Wholesale Trade, Services, and Finance.
- **Performance by sector**: Manufacturing and wholesale trade were still the best performers, with high average returns. However, manufacturing had more positive outliers over the decade.
- **Stability vs. Risk**: Finance, Insurance, and Real Estate had the most stable distributions (lowest volatility), while Mining and Services had much bigger swings from year to year.

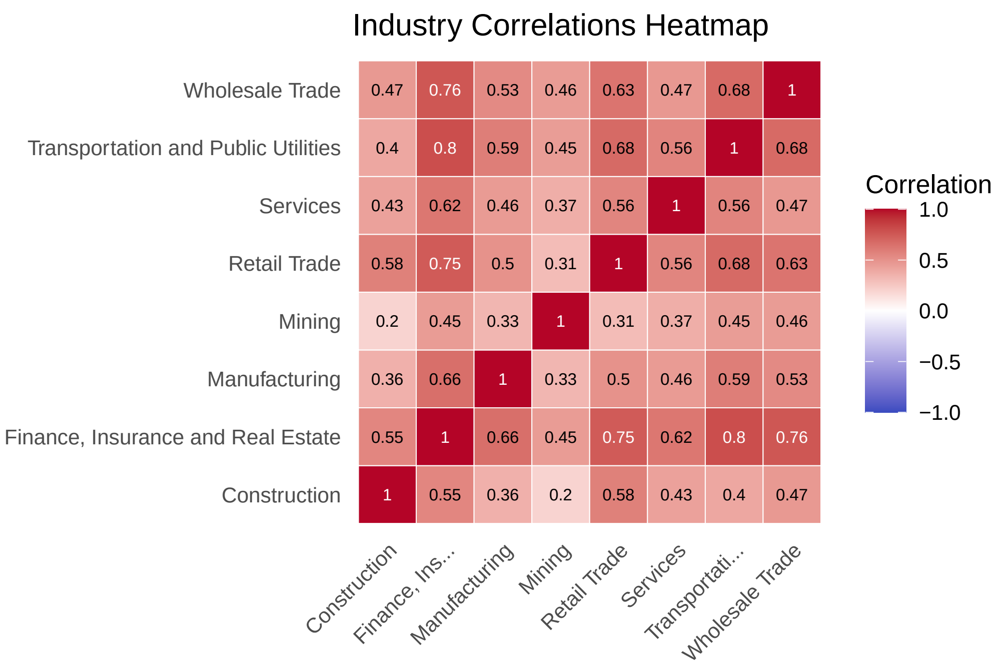

Figure 2: Industry Correlations Heatmap

Preliminary correlation analysis indicates significant systemic integration between industries. For instance, the correlation coefficient between sectors Transportation and Public Utilities (E) and Finance, Insurance and Real Estate (H) is 0.80, highlighting a shared sensitivity to market conditions. Given this high dimensionality and strong inter-variable relationship, PCA is an appropriate methodology for isolating the primary component that parallel with broad market movements.

4 PCA Preparation (100 words)

The selection of variables for PCA is limited to the 83 monthly stock return columns, as including the date or benchmark factors would bias the mathematical extraction of common variation. Since these stocks belong to diverse sectors with varying levels of price volatility, standardisation is a must. Without scaling, stocks with the highest return variance would disproportionately influence the principal components, regardless of their actual contribution to market-wide trends. By centering and scaling the data, we ensure each company contributes equally to the analysis. The PCA is then implemented on this standardised matrix to transform the correlated returns into a set of linearly uncorrelated principal components.

5 PCA and Latent Factors Analysis (800 words)

5.1 Q1 Does the largest principal component coincide with market returns?

To determine the number of components representing meaningful signals, we generate a Scree Plot. In high-dimensional financial data, we typically observe a sharp elbow after the first component, which indicates that one dominant latent factor is responsible for the majority of the co-movement across all companies.

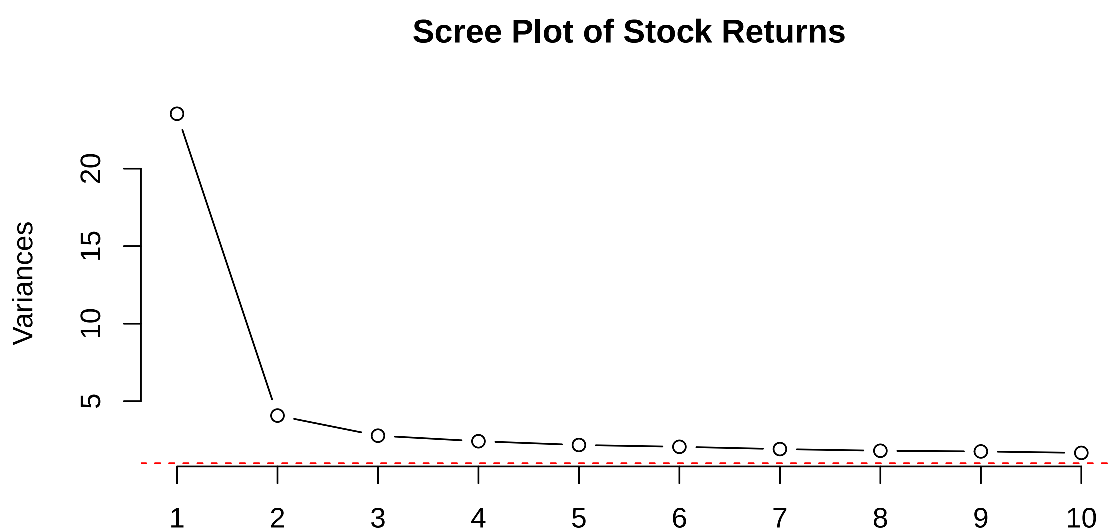

Figure 3: Scree Plot of Stock Returns

	PC1	PC2	PC3	PC4	PC5
Standard deviation	4.851678	2.018468	1.666329	1.555238	1.475485
Proportion of Variance	0.283600	0.049090	0.033450	0.029140	0.026230
Cumulative Proportion	0.283600	0.332690	0.366140	0.395280	0.421510

Scree Plot of Stock Returns

The Variance Explained by the first principal component (PC1) is a critical measure of systematic risk. If PC1 accounts for a large percentage (e.g., >30%) of the total variation in an 83-stock universe, it suggests a powerful common driver.

To test if this mathematical construct is in conformity with the actual economy, we compare the PC1 scores against the observable Market Return provided in the dataset.

```
Correlation between PC1 and Market Return: 0.9657
```

A correlation magnitude near or above 0.90 indicates compelling evidence that the first principal component (PC1) is a robust proxy for the broad market. While PCA is a purely mathematical algorithm with no prior knowledge of financial indices, the fact that its primary axis of variation aligns so closely with the broad market return confirms that the market is the single most significant latent force driving individual stock returns. Consequently, PC1 can be interpreted as the Systematic Risk Factor, representing macroeconomic events that affect the entire 83-stock universe together.

5.2 Q2 Do Market Return, SMB, and HML explain common variation in stock returns?

Table 3: Correlation between Principal Components and Market Factors

component	MarketReturn	SMB	HML
PC1	0.9657	0.4817	0.2852
PC2	0.0225	0.2410	0.4157
PC3	-0.0746	-0.0082	0.0495

To examine whether market factors explain common variation in stock returns, we combined the principal component scores with the market factors using the date variable. We then calculated correlations between the principal components and Market Return, SMB, and HML.

As shown in Table 3, PC1 has a very strong positive relationship with Market Return (0.9657), which means it captures overall market movement. PC1 also has a moderate relationship with SMB (0.4817) and a weaker relationship with HML (0.2852). PC2 has very little relationship with Market Return (0.0225), but shows a moderate relationship with HML (0.4157), suggesting it may capture value-related effects. PC3 shows very weak relationships with all three factors.

These results show that Market Return explains most of the variation in stock returns, while SMB and HML only explain a smaller part. This means the Fama-French three-factor model is only partially supported and does not fully explain the patterns found in the data.

5.3 Q3 Which industries move in the same direction as the common factors?

Table 4: Industry-level PC1 loadings

Industry	Mean PC1 Loading	SD of Loading	Min Loading	Max Loading	Number of Stocks
Finance, Insurance and Real Estate	0.118	0.050	0.005	0.199	32
Construction	0.117	NA	0.117	0.117	1
Wholesale Trade	0.107	0.021	0.086	0.144	6
Transportation and Public Utilities	0.107	0.033	0.049	0.158	12
Retail Trade	0.098	0.024	0.054	0.131	10
Manufacturing	0.082	0.042	0.031	0.144	9
Mining	0.079	0.036	0.040	0.110	4
Services	0.065	0.022	0.037	0.101	9

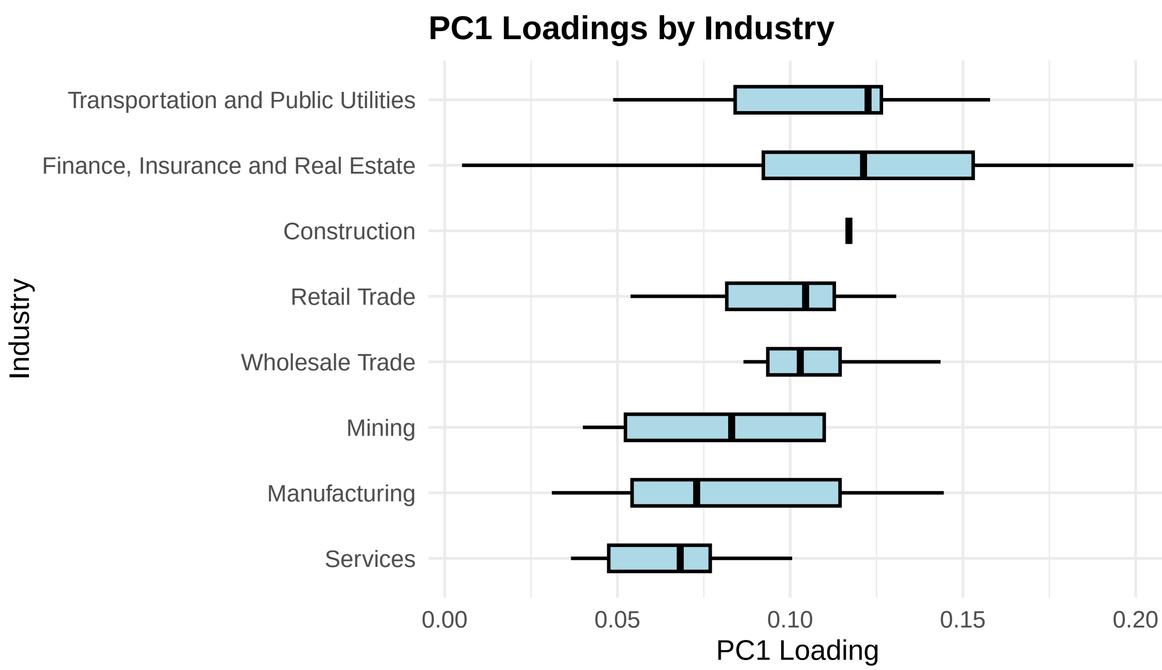

Figure 4: Distribution of PC1 loadings by industry

As shown in Table 4 and Figure 4, all industries have positive PC1 loadings, indicating that they move in the same direction as the dominant common factor. This confirms that the common factor affects the entire market rather than a specific group of industries.

However, the strength of this relationship differs across industries. Finance, Insurance and

Real Estate have the highest average loading (0.118), followed closely by Construction (0.117) and Wholesale Trade (0.107), suggesting these industries are more sensitive to the common factor. In contrast, Services (0.065) and Mining (0.079) have lower average loadings, indicating weaker exposure.

More importantly, there is clear heterogeneity within industries. Finance, Insurance and Real Estate shows a relatively large spread (SD = 0.050 and a wide range from 0.005 to 0.199), indicating that stocks within this industry respond very differently to the common factor. Manufacturing and Mining also display noticeable variation, suggesting inconsistent behaviour across stocks. In contrast, Wholesale Trade and Retail Trade show smaller spreads, indicating more homogeneous responses.

The Construction industry appears as a single line in Figure 4 and has no variation in Table 4 because it contains only one stock. As a result, it is not meaningful to assess heterogeneity within this industry.

Overall, while all industries move in the same direction as the common factor, the key result is that the degree of heterogeneity varies substantially across industries, suggesting that industry classification alone does not fully explain stock behaviour.

5.4 Q4 Which industry has the highest systematic risk?

Table 5: Correlation between Principal Components and Market Factors

Industry	No of stocks	Mean PC1 correlation	Mean PC1 variance share
Finance, Insurance and Real Estate	32	0.571	0.382
Construction	1	0.568	0.322
Transportation and Public Utilities	12	0.520	0.293
Wholesale Trade	6	0.521	0.280
Retail Trade	10	0.473	0.236
Manufacturing	9	0.396	0.194
Mining	4	0.384	0.171
Services	9	0.317	0.111

To identify the industry with the highest systematic risk, we use PC1 as the main market-wide factor. This is reasonable because the earlier result showed that PC1 moves very closely with the actual Market Return.

Based on the results on Table 5, Finance, Insurance and Real Estate appears to have the highest systematic risk. It has the highest average PC1 variance share, at 0.382, and the highest average PC1 correlation, at 0.571, across 32 stocks. This suggests that returns in this industry are more strongly driven by the shared market movement than the other industries.

While Construction also has a relatively high PC1 variance share, at 0.322, but this result comes from only one stock. Because of that, it is risky to say as a whole industry conclusion. Finance, Insurance and Real Estate is the stronger answer because it has both high exposure to PC1 and a much larger number of stocks. This also makes sense in practice, because financial firms are usually sensitive to wider market conditions, investor confidence, credit and economic conditions.

5.5 Q5 Which industry is counter-cyclical for hedging purposes?

Table 6: Industry exposure to PC1 for hedging assessment

Industry	No of stocks	Mean PC1 correlation	Mean PC1 variance share	Signal
Services	9	0.317	0.111	Weak positive exposure
Mining	4	0.384	0.171	Weak positive exposure
Manufacturing	9	0.396	0.194	Weak positive exposure
Retail Trade	10	0.473	0.236	Positive exposure
Transportation and Public Utilities	12	0.520	0.293	Positive exposure
Wholesale Trade	6	0.521	0.280	Positive exposure
Construction	1	0.568	0.322	Positive exposure
Finance, Insurance and Real Estate	32	0.571	0.382	Positive exposure

No industry in this sample as shown Table 6 is truly counter-cyclical because all industries have positive average PC1 correlations. This means they still tend to move in the same direction as the market wide factor. However, Services is the most defensive option because it has the lowest average PC1 correlation (0.317) and the lowest average PC1 variance share (0.111). This suggests that Services is the least exposed to broad market movements, although it should not be treated as a true hedge that moves opposite to the market.

6 Conclusion (150 words)

Based on the high-dimensional analysis of NYSE stock returns from 2005 to 2019, the study concludes that **a single dominant latent factor drives common market variation**.

Key Findings and Direct Answers

Q1: Market Coincidence: The first principal component (PC1) is a robust proxy for the market, showing a 0.9657 correlation with broad market returns.

Q2: Fama-French Model: While PC1 captures overall movement, SMB and HML factors explain only a smaller portion of variation, providing only partial support for the three-factor model.

Q3: Industry Direction: All analyzed industries move in the same direction as the common factor, confirmed by universally positive PC1 loadings.

Q4: Systematic Risk: Finance, Insurance and Real Estate carries the highest systematic risk, with the highest PC1 correlation (0.571) and variance share (0.382).

Q5: Hedging: No industry is truly counter-cyclical; however, Services is the most defensive option due to its minimal market exposure.

Investment Implications

Investors should recognize that diversification across these industries does not eliminate systematic risk, as all sectors remain positively correlated with the market. For hedging, the Services sector provides the best relative protection. Conversely, the Finance sector requires careful monitoring during macroeconomic shifts due to its high sensitivity to latent market forces.

7 Reference

8 Appendix

8.1 Data and Preparation Code

```r
# 1. Load Libraries
library(tidyverse)
library(janitor)
library(psych)
library(knitr)
library(stringr)
```

```r
library(scales)

# 2. Load Data
stocks <- read_csv("SampleH.csv")
market_factors <- read_csv("MarketFactors.csv")

# 3. Define the mapping of letters to Industry Names
industry_lookup <- c(
  "B" = "Mining",
  "C" = "Construction",
  "D" = "Manufacturing",
  "E" = "Transportation and Public Utilities",
  "F" = "Wholesale Trade",
  "G" = "Retail Trade",
  "H" = "Finance, Insurance and Real Estate",
  "I" = "Services"
  )

# 4. Transform and Merge Data
# We pivot the stocks to a 'long' format so each row is a unique Stock-Date observation
data_complete <- stocks |>
    pivot_longer(
        # Keep Date as is
        cols = -Date,
        # New column for stock codes
        names_to = "stock_id",
        # new column for the return values
        values_to = "return"
        ) |>
        mutate(industry_letter = substr(stock_id, 1, 1)) |>
        mutate(industry_name = industry_lookup[industry_letter]) |>
        inner_join(market_factors, by = "Date") |>
        mutate(year = substr(Date, 2, 5)) |>
        # Reorder: put `Return` column after `industry_name`
        relocate(return, .after = industry_name) |>
        relocate(year, .before = Date) |>
        # clean names
        clean_names()

kable(head(data_complete))

# 5. Check dimensions and missing values
```

```
explain_dim <- function(df) {
  dimensions <- dim(df)
  cat("The dataset '", substitute(df), "' contains:\n", sep = "")
  cat("- ", dimensions[1], " observations (rows)\n", sep = "")
  cat("- ", dimensions[2], " variables (columns)\n", sep = "")
}
explain_dim(data_complete)
cat("Total amoun of NA entries:", sum(is.na(data_complete)))
```

8.2 Preliminary Analysis Code

Table group statistics by the 'industries'

```
# 1. Group statistics by the 'industries' vector created earlier
eda_summary <- data_complete |>
    group_by(industry_name) |>
    summarise(
        avg_return_monthly = label_percent(accuracy = 0.01)(mean(return, na.rm = TRUE)),
        volatility = label_percent(accuracy = 0.01)(sd(return, na.rm = TRUE)),
        num_stocks = n_distinct(stock_id)
    ) |>
      arrange(desc(avg_return_monthly))
kable(eda_summary)

# create the ordered vector
industry_order <- eda_summary$industry_name
```

Time-series boxplots

```
# 2. Create the Yearly Boxplot for 2 industries per plot
data_complete_group <- data_complete|>
  mutate(industry_name = factor(industry_name, levels = industry_order)) |>
  mutate(plot_group = as.numeric(factor(industry_name)),
         plot_group = ceiling(plot_group / 2))

# We facet by industry_name to see the movement of each sector clearly
data_complete_group |>
  filter(plot_group == 1) |>
  ggplot(aes(x = factor(year), y = return, fill = industry_name)) +
  geom_boxplot(outlier.size = 0.5,
                alpha = 0.7,
```

```r
               width = 0.7,
               median.linewidth = 1) +
  facet_wrap(~industry_name,
             scales = "free_y",
             nrow = 2,
             labeller = label_wrap_gen(width = 20)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50", alpha = 0.5) +
  theme_minimal() +
  labs(
    title = "Industry Return Distributions: 2005 - 2019",
    subtitle = "Boxplots show median, quartiles, and volatility shifts per year",
    x = "Year",
    y = "Monthly Return (%)"
  ) +
  theme(
    # Reduce x-axis clutter: only show every 3rd year
    axis.text.x = element_text(angle = 45, hjust = 1, size = 8),
    strip.text = element_text(face = "bold", size = 10),
    panel.spacing = unit(2, "lines"),
    legend.position = "none"
  ) +
    scale_x_discrete(breaks = seq(2005, 2019, by = 2))

data_complete_group |>
  filter(plot_group == 2) |>
  ggplot(aes(x = factor(year), y = return, fill = industry_name)) +
  geom_boxplot(outlier.size = 0.5,
               alpha = 0.7,
               width = 0.7,
               median.linewidth = 1) +
  facet_wrap(~industry_name,
             scales = "free_y",
             nrow = 2,
             labeller = label_wrap_gen(width = 20)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50", alpha = 0.5) +
  theme_minimal() +
  labs(
    x = "Year",
    y = "Monthly Return (%)"
  ) +
  theme(
    # Reduce x-axis clutter: only show every 3rd year
```

```r
    axis.text.x = element_text(angle = 45, hjust = 1, size = 8),
    strip.text = element_text(face = "bold", size = 10),
    panel.spacing = unit(2, "lines"),
    legend.position = "none"
  ) +
    scale_x_discrete(breaks = seq(2005, 2019, by = 2))

data_complete_group |>
  filter(plot_group == 3) |>
  ggplot(aes(x = factor(year), y = return, fill = industry_name)) +
  geom_boxplot(outlier.size = 0.5,
              alpha = 0.7,
              width = 0.7,
              median.linewidth = 1) +
  facet_wrap(~industry_name,
            scales = "free_y",
            nrow = 2,
            labeller = label_wrap_gen(width = 20)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50", alpha = 0.5) +
  theme_minimal() +
  labs(
    x = "Year",
    y = "Monthly Return (%)"
  ) +
  theme(
    # Reduce x-axis clutter: only show every 3rd year
    axis.text.x = element_text(angle = 45, hjust = 1, size = 8),
    strip.text = element_text(face = "bold", size = 10),
    panel.spacing = unit(2, "lines"),
    legend.position = "none"
  ) +
    scale_x_discrete(breaks = seq(2005, 2019, by = 2))

data_complete_group |>
  filter(plot_group == 4) |>
  ggplot(aes(x = factor(year), y = return, fill = industry_name)) +
  geom_boxplot(outlier.size = 0.5,
              alpha = 0.7,
              width = 0.7,
              median.linewidth = 1) +
  facet_wrap(~industry_name,
            scales = "free_y",
```

```r
              nrow = 2,
              labeller = label_wrap_gen(width = 20)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50", alpha = 0.5) +
  theme_minimal() +
  labs(
    x = "Year",
    y = "Monthly Return (%)"
  ) +
  theme(
    # Reduce x-axis clutter: only show every 3rd year
    axis.text.x = element_text(angle = 45, hjust = 1, size = 8),
    strip.text = element_text(face = "bold", size = 10),
    panel.spacing = unit(2, "lines"),
    legend.position = "none"
  ) +
    scale_x_discrete(breaks = seq(2005, 2019, by = 2))
```

Heatmap plot

```r
# 3. Calculate Average Monthly Returns per Industry (time series) for correlation analysis
industry_ts <- data_complete |>
    # Group by Industry and Date
    group_by(industry_name, date) |>
    # Calculate the average return for the each industry classification for that month
    summarise(avg_return = mean(return, na.rm = TRUE), .groups = "drop") |>
    # Pivot so each industry gets its own column
    pivot_wider(names_from = industry_name, values_from = avg_return)

# 4. Create Correlation Matrix between Industry Averages
ind_cor_long <- industry_ts |>
    select(-date) |>
    cor(use = "complete.obs") |>
    as.table() |>
    as.data.frame()

# 5. Use heatmap to visualise
ggplot(ind_cor_long, aes(Var1, Var2, fill = Freq)) +
  geom_tile(color = "white") +
  geom_text(aes(label = round(Freq, 2)),
            color = ifelse(abs(ind_cor_long$Freq) > 0.7,"white", "black"), size = 2.5) +
  scale_fill_gradient2(low = "#3B4CC0", high = "#B40427", mid = "white",
                       midpoint = 0, limit = c(-1,1), name="Correlation") +
```

```r
  coord_fixed() +
  theme_minimal() +
  labs(title = "Industry Correlations Heatmap", x = NULL, y = NULL) +
  scale_x_discrete(labels = function(x) str_trunc(x, 15)) +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1),
        panel.grid.major = element_blank(), # Remove background grid
        legend.position = "right"
  )
```

8.3 PCA Preparation Code

```r
# 1. Select only stock return columns (numeric variables)
# We exclude 'date' and the Fama-French factors
pca_input <- stocks |> select(-Date)

# 2. Standardise and Run PCA
# 'scale. = TRUE' ensures data is standardised (mean=0, sd=1)
pca_res <- prcomp(pca_input, scale. = TRUE)
```

8.4 Q1 Code

Scree Plot

```r
# 1. Visualize Eigenvalues
screeplot(pca_res, type = "lines", main = "Scree Plot of Stock Returns")
abline(h = 1, col="red", lty=2) # Kaiser Criterion threshold

# 2. Calculate Variance Explained
kable(summary(pca_res)$importance[, 1:5])
```

Comparison PC1 with Market Return

```r
# 1. Extract PC1 scores
pc1_scores <- pca_res$x[, 1]

# 2. Correlate with Market Return
market_test_corr <- cor(pc1_scores, market_factors$MarketReturn)

cat("Correlation between PC1 and Market Return:", round(market_test_corr, 4))
```

8.5 Q2 Code

Table: Correlation between Principal Components and Market Factors

```r
# Extract PCA scores
pc_scores <- as.data.frame(pca_res$x) |>
  mutate(Date = stocks$Date)

# Join PCA scores with market factors by Date
q2_data <- pc_scores |>
  inner_join(market_factors, by = "Date")

# Correlation table between first 3 PCs and market factors
q2_cor_table <- q2_data |>
  select(PC1, PC2, PC3, MarketReturn, SMB, HML) |>
  cor(use = "complete.obs") |>
  as.data.frame() |>
  rownames_to_column("component") |>
  filter(component %in% c("PC1", "PC2", "PC3")) |>
  select(component, MarketReturn, SMB, HML) |>
  mutate(across(where(is.numeric), ~ round(.x, 4)))

kable(q2_cor_table)
```

8.6 Q3 Code

Table: Industry-level PC1 loadings

```r
pc1_loadings |>
  ggplot(aes(x = reorder(industry, pc1_loading, median), y = pc1_loading)) +
  geom_boxplot(fill = "lightblue", color = "black", width = 0.4, size = 0.7) +
  coord_flip() +
  labs(
    title = "PC1 Loadings by Industry",
    x = "Industry",
    y = "PC1 Loading"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold"),
    axis.text.y = element_text(size = 10),
```

```r
    axis.text.x = element_text(size = 10)
  )
```

Boxplot: Distribution of PC1 loadings by industry

```r
pc1_loadings |>
  ggplot(aes(x = reorder(industry, pc1_loading, median), y = pc1_loading)) +
  geom_boxplot(fill = "lightblue", color = "black", width = 0.4, size = 0.7) +
  coord_flip() +
  labs(
    title = "PC1 Loadings by Industry",
    x = "Industry",
    y = "PC1 Loading"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(face = "bold"),
    axis.text.y = element_text(size = 10),
    axis.text.x = element_text(size = 10)
  )
```

8.7 Q4 Code

Industry risk based on PC1

```r
pc1_sign <- ifelse(
  cor(pca_res$x[, 1], market_factors$MarketReturn, use = "complete.obs") < 0,
  -1,
  1
)

# Stock exposure to PC1
stock_pc1 <- tibble(
  stock_id = colnames(pca_input),
  industry_letter = substr(stock_id, 1, 1),
  industry_name = as.character(industry_lookup[industry_letter]),
  pc1_loading = pc1_sign * pca_res$rotation[, 1],
  pc1_corr = pc1_loading * pca_res$sdev[1],
  pc1_variance_share = pc1_corr^2
)
```

```
q4_industry_risk <- stock_pc1 |>
  group_by(industry_name) |>
  summarise(
    n_stocks = n(),
    mean_pc1_corr = mean(pc1_corr, na.rm = TRUE),
    mean_pc1_variance_share = mean(pc1_variance_share, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(desc(mean_pc1_variance_share)) |>
  mutate(across(-c(industry_name, n_stocks), ~ round(.x, 3))) |>
  rename(
    Industry = industry_name,
    `No of stocks` = n_stocks,
    `Mean PC1 correlation` = mean_pc1_corr,
    `Mean PC1 variance share` = mean_pc1_variance_share
  )

kable(q4_industry_risk)
```

8.8 Q5 Code

Industry exposure to PC1

```
q5_hedging <- stock_pc1|>
  group_by(industry_name)|>
  summarise(
    n_stocks = n(),
    mean_pc1_corr = mean(pc1_corr, na.rm = TRUE),
    mean_pc1_variance_share = mean(pc1_variance_share, na.rm = TRUE),
    .groups = "drop"
  ) |>
  arrange(mean_pc1_corr)|>
  mutate(
    signal = case_when(
      mean_pc1_corr < 0 ~ "Counter-cyclical",
      mean_pc1_corr < 0.40 ~ "Weak positive exposure",
      TRUE ~ "Positive exposure"
    )
  )|>
  mutate(across(-c(industry_name, n_stocks, signal), ~ round(.x, 3))) |>
  rename(
```

```
    Industry = industry_name,
    `No of stocks` = n_stocks,
    `Mean PC1 correlation` = mean_pc1_corr,
    `Mean PC1 variance share` = mean_pc1_variance_share,
    `Signal` = signal
  )

kable(q5_hedging)
```